

Physics105 Common Exam 1 : Fall 2010

Solution

B

Name (Print): _____ Digit ID: _____ Section: _____

Honors Code Pledge: As an NJIT student I _____, pledge to comply with the provisions of the NJIT Academic Honor Code. I assert that I have not violated the NJIT Academic Honor Code.

Instructions:

There are 10 multiple choice questions on this exam. They all have the same value (10 points).

Answer each question on the Scantron sheet using a #2 pencil. There is no penalty for guessing. You will need to do calculations on the exam papers for most of the questions. Use the backs of the exam papers for extra space.

Be sure your name and section number are on both the Scantron form and the exam book.

The formula sheet is provided.

Make sure to bring your own calculator. Sharing of calculators is not permitted.

As you know, NJIT has a zero-tolerance policy for ethics code violations. Students are not to communicate with each other once the exam has started. All cell phones, pagers, or similar devices must be turned off.

If you have questions or need something, call your proctor or instructor.

1. A standard exam page is 8.5 inches by 11.0 inches. Its area in cm^2 is

(1 inch = 2.54cm)

A. 15

B. 37

C. 94

D. 237

E. 603

$$8.5 \text{ in} = 8.5 \times 2.54 \text{ cm} = 21.6 \text{ cm}$$

$$11.0 \text{ in} = 11 \times 2.54 \text{ cm} = 27.9 \text{ cm}$$

$$(\text{Area}) = 21.6 \text{ cm} \times 27.9 \text{ cm} = 602.6 \text{ cm}^2 \approx 603 \text{ cm}^2$$

2. If 'a' is the acceleration, 'v' is the velocity, 'x' is the position and 't' is the time, then, which equation is not dimensionally correct?

A. $t = x/v$

B. $a = v^2/x$

C. $v = a/t$

D. $t^2 = 2x/a$

E. None of the above

$$[t] = T, [x/v] = [x]/[v] = L/(L/T) = T$$

$$[a] = L/T^2, [v^2/x] = (L/T)^2/L = L/T^2$$

$$[v] = L/T, [a/t] = (L/T^2)/T = L/T^3$$

$$[t^2] = T^2, [2x/a] = L/(L/T^2) = T^2$$

3. A projectile is thrown from the top of a building with an initial velocity of 30 m/s in the horizontal direction. If the top of the building is 30 m above the ground, how fast will the projectile be moving just before it strikes the ground?

- A. 35 m/s
 B. 39 m/s
 C. 31 m/s
 D. 43 m/s
 E. 54 m/s

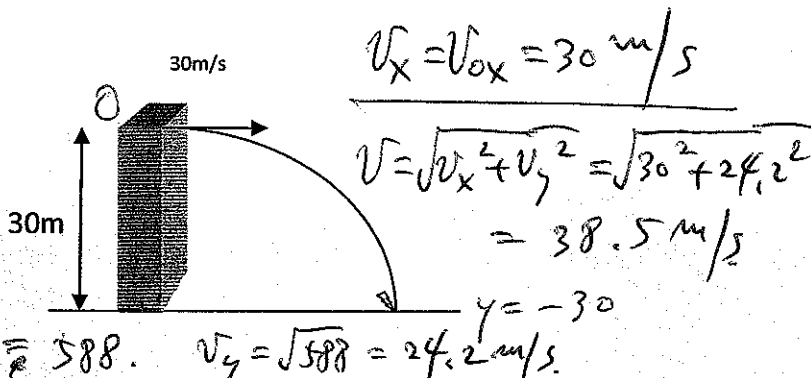
$$v_y = v_{y0} - gt$$

$$= 0 - 9.8 \times 2.4$$

$$v_y^2 = v_{y0}^2 - 2g\Delta y$$

$$= 0 - 2 \times 9.8 \times (-30) = 588$$

$$v_y = \sqrt{588} = 24.2 \text{ m/s}$$



4. An automobile moving along a straight track changes its velocity from 40 m/s to 80 m/s in a distance of 200m. What is the (constant) acceleration of the vehicle during this time?

- A. 8.0 m/s²
 B. 9.6 m/s²
 C. 12.0 m/s²
 D. 6.9 m/s²
 E. 0.20 m/s²

$$v^2 = v_0^2 + 2a\Delta x$$

$$80^2 = 40^2 + 2 \times a \times 200$$

$$\therefore 400a = 80^2 - 40^2 \therefore a = \frac{80^2 - 40^2}{400} = 12$$

5. A ball thrown vertically from ground level is caught 3.0 sec later by a person on a balcony which is 14 m above the ground. Determine the initial speed of the ball

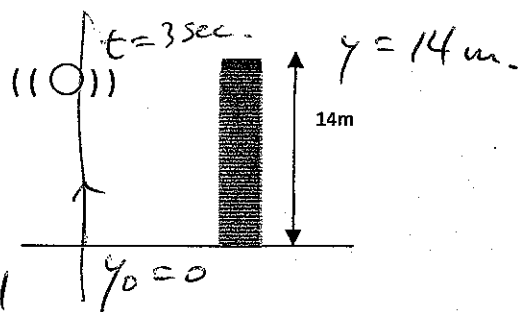
- A. 19 m/s
 B. 4.7 m/s
 C. 10 m/s
 D. 34 m/s
 E. 17 m/s

$$y = y_0 + v_0 t - \frac{1}{2} g t^2$$

$$14 = 0 + v_0 \times 3 - \frac{1}{2} \times 9.8 \times 3^2$$

$$v_0 \times 3 = 14 + \frac{1}{2} \times 9.8 \times 3^2 = 58.1$$

$$v_0 = 58.1 / 3 = 19.4 \text{ m/s}$$



6. A skier leaves a ski jump with a horizontal velocity of 29.4 m/s. The instant before she lands three seconds later, the magnitudes of the horizontal and vertical components of her velocity are:

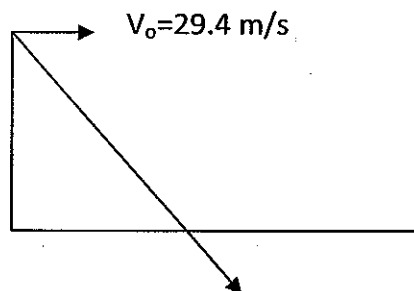
- A. 0; 29.4 m/s
 B. 29.4 m/s; 0
 C. 29.4 m/s; 29.4 m/s
 D. 29.4 m/s; 41.6 m/s
 E. 41.6 m/s; 41.6 m/s

$$v_{0x} = 29.4 = v_x$$

$$v_y = v_{y0} - gt$$

$$= 0 - 9.8 \times 3$$

$$= -29.4$$



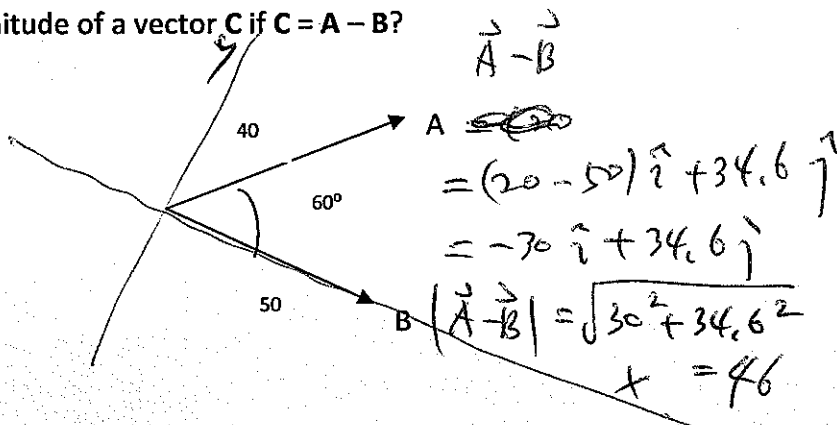
7. Vectors A and B are shown. What is the magnitude of a vector C if $C = A - B$?

- A. 46
B. 10
C. 30
D. 78
E. 90

$$\vec{B} = 50 \hat{i}$$

$$\vec{A} = 40 \cos 60^\circ \hat{i} + 40 \sin 60^\circ \hat{j}$$

$$= 20 \hat{i} + 34.6 \hat{j}$$



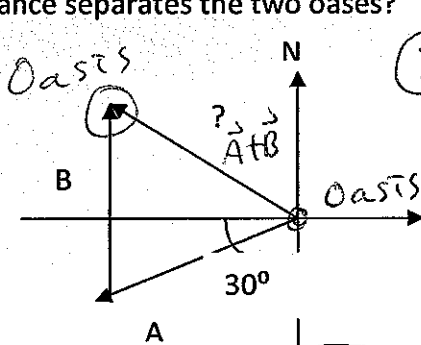
8. Starting from one oasis, a camel walks 25 km in a direction 30° south of west and then walks 30 km toward the north to a second oasis. What distance separates the two oases?

- A. 15 km
B. 48 km
C. 28 km
D. 53 km
E. 55 km

①

$$\vec{A} + \vec{B} = ?$$

$$\vec{A} = -25 \cos 30^\circ \hat{i} - 25 \sin 30^\circ \hat{j} = -21.6 \hat{i} - 12.5 \hat{j}$$



$$\vec{A} + \vec{B} = -21.6 \hat{i} + (-12.5 + 30) \hat{j}$$

$$= -21.6 \hat{i} + 17.5 \hat{j}$$

$$|\vec{A} + \vec{B}| = \sqrt{21.6^2 + 17.5^2} = 27.8$$

9. A toy rocket, launched from the ground, rises vertically with an acceleration of 20 m/s^2 for 6.0 sec until the motor stops. Disregarding any air resistance, what maximum height above the ground will the rocket achieve? $\Delta y'$

- A. 1.1 km
B. 2.7 km
C. 1.9 km
D. 0.39 km
E. 1.5 km

Before 6 sec, NOT Free Fall motion (because of motor) $a = 20$

After 6 sec. Free fall motion $a = -g = -9.8$

At 6 sec

$$y = y_0 + v_{0y}t + \frac{1}{2}at^2$$

$$= 0 + 0 + \frac{1}{2}20 \times 6^2 = 360$$

$$v_y = v_{0y} + at = 0 + 20 \times 6 = 120$$

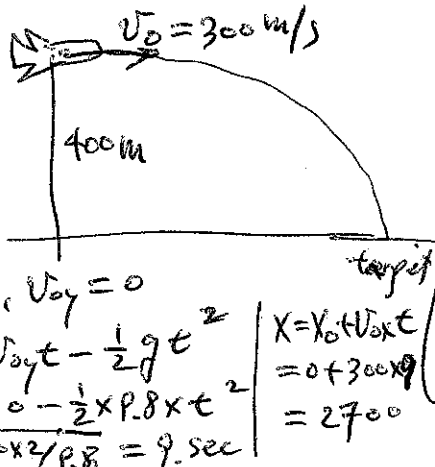
$t = 6 \text{ sec}$, motor stops

$a = 20 \text{ m/s}^2$: motion with constant acceleration.

$v_{y0} = 0$ $t = 0$

10. An airplane flies horizontally with a speed of 300 m/s at an altitude of 400m. Assume that the ground is level. At what horizontal distance from a target must the pilot release a bomb so as to hit the target?

- A. 3.0 km
B. 2.4 km
C. 3.3 km
D. 2.7 km
E. 1.7 km



Between $t = 6 \text{ sec}$ & max height,

$$v_{\text{max height}}^2 = v_{6 \text{ sec}}^2 - 2g \Delta y'$$

$$\Delta y' = \frac{120^2}{2 \times 9.8} = 735$$

$$h_{\text{total}} = \Delta y + \Delta y' = 360 + 735 = 1095 \text{ m}$$

measured from $t = 6 \text{ sec}$